Social Influence and Participation: How Socially Shared Content Boosts Civic Participation

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Abstract

Public opinion and political engagement play a critical role in the democratic process, shaping election outcomes and influencing the policies that elected officials enact. The way that individuals form political preferences and act depends on pre-existing knowledge, habits, and exposure to political information. Because all of these processes depend on each other and interact dynamically over time, it is difficult to examine how *exogenous* sources of political information can cause changes in mass opinion and behavior. Common approaches include case studies, correlational analyses of survey data, and short-term lab experiments that seldom resemble real world media exposure and often rely on captive audiences. We present an analysis of a 72-day product test at Facebook that increased exposure to socially shared political content (based on random assignment) and analyze survey data from 12.3 thousand respondents. The data allow for an externally and ecologically valid, causal examination of the effects of encountering more socially shared content related to political topics, obtained utilizing supervised text classification. We show that seeing additional political content shared by social contacts increases voter turnout and strengthens policy preferences among nonpartisans.

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> "Had Lazarsfeld remained in the field of political communication, he might have turned his attention from the immediate circle of the individual voter... to the larger public sphere in which these social and political molecules are embedded. The combined methodologies of decision and diffusion might help to explain the dynamics of deliberation and aggregation, where the media come in, and how binding social norms emerge."
>
> –Elihu Katz, 2005, in Introduction to the Transaction Edition, *Personal Influence: The Part Played by People in the Flow of Mass Communications*.

Scholars have long sought to understand how political information that individuals encounter affects political behavior. Early observational work concluded that "opinion leaders" and "creative elites" helped shaped mass opinion and prompted civic engagement (Katz and Lazarsfeld 1955; Converse 1962). Yet, those who are most likely to attend to and absorb this information tend to have much higher levels of background knowledge in the first place (Zaller 1991, 1992; Price and Zaller 1993). And it is precisely these people who tend to get less information—namely, nonpartisans—who tend to hold the "balance of power" in elections. On the other hand, less-informed individuals may be able to absorb some minimal amount of information necessary to participate effectively in the democratic process, via cues from elites (Popkin 1994; Lupia and McCubbins 1998) and social networks (Ryan 2011; Sokhey and McClurg 2012). Yet all of these processes depend on each other and interact dynamically over time—making it tremendously challenging to credibly answer the following question: How do *exogenous* sources of information and influence, such as elite discourse encountered in media, affect opinion and behavior (e.g., Siegel 2013)?

Below, we utilize an exogenous source of political information, introduced in the context of testing a proposed product change at Facebook, to identify the effect of socially shared political content on policy preferences and voter turnout. We first review the literature on the relationship between political information and political outcomes, and on newer work suggesting that individuals should be more likely attend to and deeply process socially shared content. Then, we describe our measure of exposure to political information, outline the design of our instrumental variables design, and present our analysis strategy. Lastly, we present a series of results based on a survey issued during the 12 days following the 2012 election, to which 12.3 thousand participants responded. This analysis shows that additional exposure to socially shared political content strengthened policy preferences among independents and increased self-reported turnout.

Political information, opinion and participation

Because exposure to political information is endogenous with background political knowledge, partisan identification, and one's network of social contacts, it is difficult to identify the effect of political information using observational approaches. The most common approaches comprise of case studies and correlational analyses of survey data, which do not allow us to examine counterfactual questions of how individuals who would not have been exposed to media respond. Other approaches comprises of short-term lab experiments with precise counterfactuals, but that seldom resemble real world media exposure in terms of duration, context, choice, and level of attention devoted to media consumption (Hovland 1959; Druckman et al. 2012, critiques of the "captive audience" assumption in experiments are not new but only recently have begun to be seriously addressed in the literature).

Nonetheless, evidence and expectations about the effects of media exposure continue to accumulate. Despite early evidence of the "minimal effects" of media on political outcomes (Klapper 1960), an array of more recent work has established expectations that political information should affect policy preferences and behavior through knowledge (Althaus 1998; Lau and Redlawsk 1997), or through impressions that are updated continuously (even if that information is not remembered Lodge et al. 1989, 1995). There is also observational evidence that elite discourse can more directly influence behavior, increasing (or in some cases decreasing) one's level of political participation (e.g., Ansolabehere et al. 1991; George and Waldfogel 2006; DellaVigna and Kaplan 2007; Chiang and Knight 2011).

Some of the best evidence to date of the effects of elite discourse come from natural or quasi-experiments. For example, the introduction of television is associated with a decline in newspaper reading and voter turnout (newspaper consumption being associated with political participation, Gentzkow 2006). Other work finds an association between the displacement of local papers by the *New York Times* and a decline in turnout in nonpresidential elections relative to presidential elections (George and Waldfogel 2006). And, Ladd and Lenz (2009) use a rare reversal in British newspaper endorsements to document the effect of elite endorsements on vote choice.

There is also a long tradition of showing how media can affect opinion via framing experiments that take place with lab controls. This work usually involves single-shot lab experiments that show that people adjust their attitudes in the direction of the frame to which they are exposed (e.g., when presented with arguments that estate taxes are "double taxation," people grow increasingly opposed to it, see Chong and Druckman 2007). Of course, opinion forms as part of a dynamic, over-time process. Indeed, studies that expose individuals to competing frames simultaneously tend to find a canceling-out effect, and those that expose individuals to a series of frames over time show that framing effects tend to decay

among those who do not carefully process and absorb these messages(Chong and Druckman 2010; Druckman et al. 2012). Furthermore, initial work using a far more realistic design—in which people are given a choice of consuming a wide array of framing messages or not, over time—shows that people motivated to process the initial message are more likely to accept its argument and then disregard later, competing frames (Druckman et al. 2012). Motivated and interested individuals tend to form an opinion after exposure to an initial message then search out consistent messages (Mutz 2006; Iyengar et al. 2008; Iyengar and Hahn 2009; Kim et al. 2010; Messing and Westwood 2012).

Choice and selectivity are more important today than ever. The wide array of entertainment options (Prior 2007), topically specialized publications, and openly partisan sources (Bennett and Iyengar 2008) have lead to concerns about polarization and political ignorance in the mass public, as the majority of news consumers "opt out" of news in favor of entertainment alternatives, while partisan consumers habituate themselves to partisan news outlets (Mutz 2006; Iyengar et al. 2008; Iyengar and Hahn 2009). Of course, these questions matter even more today with the vast choice available to consumers thanks to the current fragmented and ubiquitous media environment. Indeed, it is far more productive to think about an *individual agenda* rather than a national agenda when considering what is likely to drive political behavior.

How does this dynamic process coalesce to affect opinions and behavior over time? The one publicly available field experiment examining this question has found mostly null effects—or if anything a subtle shift to the left. Gerber et al. (2009) randomly assigning a subscription to the Washington Post or the Washington Times for ten weeks. They found that individuals who received *either paper* were more likely to vote for the Democratic candidate, but found no effect on political knowledge, stated opinions, or turnout in post-election survey and voter data. The authors do note "Some evidence from voting records also suggests that receiving either paper led to increased 2006 voter turnout," however, leaving us some room for optimism.

Mass media and social networks

Often social interaction and media are conceptualized as separate and competing influences on public opinion and behavior. The studies that examine both phenomena simultaneously suggest that elite discourse in media and social influence from peers act as "substitutes"—the greater the influence from one's social network on preferences and behavior, the less the media can affect these outcomes (Mondak 1995; Beck et al. 2002; Druckman and Nelson 2003; Siegel 2013). Indeed, the mass adoption of television in the 1960s and

1970s is associated with declining community involvement (see for example Putnam 2001), a phenomena hypothesized to limit the role of "opinion leaders" in transmitting opinion and viewpoints expressed in news media to the public at large via interpersonal discussion (Katz and Lazarsfeld 1955). Political models of mass opinion describing this media regime, while acknowledging the importance of pre-existing knowledge and habitual exposure to elite opinion in the mass media, assumed that exposure was uniform—largely independent of social influence, partisanship, and other social cleavages.(e.g., Zaller 1991, 1992).[1]

More recently, it has also been hypothesized that social network structure *conditions* the media's impact on voter turnout and public opinion (Siegel 2013). The beliefs and behavior of others in one's social network affect one's own beliefs and behavior via direct influence (e.g., via the provision of social capital, Chong 1991; La Due Lake and Huckfeldt 1998), and via the provision of information (especially for the uninformed, Ryan 2011; Sokhey and McClurg 2012).

Indeed, with the vast choice set available online, the social provisioning of information can serve to alter the set of political information to which an individual is exposed. As DiMaggio et al. (2004) point out, with effort, one can find virtually anything on the internet, but the ease of doing so shapes exposure and reception. As email and social media technologies have gained in popularity, sharing content with one's social contacts has become increasingly easy and common.[2] Recent literature shows that social factors shape attention to content to an even stronger degree than ideological selective exposure based on media source brands. In particular, individuals are more likely to attend to messages when endorsed by multiple peers (Messing and Westwood 2012), when shared by a close social tie (Messing and Westwood 2013), and/or when shared by another in-group member (Messing 2013).

A large literature also suggests that social influence should not merely increase the likelihood that an individual receives and attends to content, but should also shape the way that an individual processes that message and whether she ultimately adjusts her opinions and behavior in response. The social psychology literature shows that delivery of a message by a social contact increases the likelihood that an individual will engage deeply and makes persuasion more likely (by increasing both personal relevance and motivation to process, e.g., Sherif et al. 1965; Petty and Cacioppo 1979; Axsom et al. 1987; Brinol and Petty 2009).

At the same time, there are a number of reasons that content encountered in social me-

[1] Of course, this "two-message model" and other similar models do of course acknowledge that *opinion change* depends on pre-existing viewpoints and the ability and context to process elite opinion.

[2] As of 2011, Facebook was among the top sources of referral traffic to the web's top traditional websites (data from Nielsen's Netview database, see Pew 2011); Twitter was a significant source as well. By 2012, 42 percent of Americans (not just social media users) reported learning something about the 2012 campaign on Facebook, according to Pew (2012).

dia is increasingly likely to contain persuasive political messages—elite opinion and advocacy content. Work on "social transmission" has found that people tend to share the most arousing, emotionally engaging, opinion-laden, and persuasive messages they encounter (Berger and Milkman 2012, indeed our own re-analysis of this data shows that content from the opinion section is most likely to be shared, see Appendix I). Furthermore, strategic political entities have attempted to use these communication technologies to "energize the base"—political parties and activists now frequently attempt to create compelling arguments, then call on the party faithful to spread this content to their contacts via email and social media (and especially so on the political left, see, Kreiss 2012).[3]

The literature provides clear reasons to expect that socially shared political content can shape political preferences and behavior and play an important role in informing the public, yet the study of its effects has proven extremely difficult. Below, we examine how socially sharing opinion content affects how people consume political information, the extent to which they hold strong political opinions, and how this can affect actual political behaviors including turnout. First, however, we outline our method to measure exposure to political content on Facebook.

Measure of exposure to shared political content

We begin by describing our measure of exposure, which we will use both when providing empirical context and when describing our causal estimates. To produce such a measure, we classify the news media content that participants' friends' shared from the beginning of 2012 to election day.

During the 72-day study period, participants' friends shared 9.8 million URLs, 6.5 million of which were unique.[4] These items appeared in participants' news feeds, which displayed a photograph, story title, domain, and a two to three sentence summary associated with the item (see Figure 5), which are produced by the originating website. We classify the text of each of these summaries as political or not, then determine which items appeared in each participant's news feed using exposure-logging data.

Our classification strategy is to use the information embedded in URLs as labels indicative of political content, then build a model to identify words and phrases predictive of content

[3]These parties and activist groups, including Upworthy (created by MoveOn.org's Eli Pariser and Peter Koechley formerly of The Onion), have assembled massive contact lists in the course of shifting fundraising strategies toward targeting many individual donors, largely in response to the McCain-Feingold campaign finance reform caps on individual donations and changing campaign strategies that increasingly rely on get out the vote campaigns targeting individuals (Gerber and Green 2000; Gerber et al. 2008, based on discoveries about the cost-effectiveness of stimulating turnout among supporters, e.g.,[]).

[4]For the purpose of this analysis, we consider only those participants who completed the survey.

related to politics during the period under study. Once we have a model of these words and phrases, we can use it to classify unseen texts (commonly known as supervised machine learning, see for example, Hillard et al. 2008; Hopkins and King 2010). We label as "political content" all items that originated from our list of news domains *and* whose urls contained the tokens "politi," "usnews," "national," "election," "obama," and "romney." We then took content whose URLs contained variants of "sports," "entertainment," "arts," "style"[5] (from any URL) and label these items as "non-political content."[6] This training corpus of labeled item summaries consists of 102,651 documents—77,866 non-political items and 24,785 political items. For each of these documents, we then computed unigram, bigram, and trigram features—counts of single words, word pairs, and word triplets, respectively.[7] Next, we remove features that occurred in more than half of all documents, or in only a single document, leaving 348,643 features. We represent each document (rows) and each feature (columns) in a matrix, X, which we represent in compressed sparse row (CSR) format.

Finally we fit a linear Support Vector Machine (SVM) classifier to the data. SVMs find the vector that maximizes separation between classes in multiple dimensions. We rely on the hinge loss function and fit the model via coordinate descent, implemented in the Python package "Scikit-Learn," (Pedregosa et al. 2011) which uses on the C++ "liblinear" library (Fan et al. 2008). Formally, the model fits the following function to the data:

$$f(x) = \sum_{j=1}^{d} h(\mathbf{x})^T \beta + \beta_0$$

The model is fit by solving an optimization problem that can be written as minimizing the following hinge loss function with penalty:

$$\min_{\beta_0,\beta} \sum [1 - y_i f(\mathbf{x}_i)]_+ + \frac{\lambda}{2}||\beta||^2$$

Where the data comprise $\mathbf{x}_i \in \mathbb{R}^p, y_i \in -1, 1$, the subscript $+$ corresponds to the positive part, and λ is equivalent to the inverse of the cost parameter, C, which regulates the models' sensitivity to non-separable points (Hastie et al. 2009). Here we merely use our features as

[5]The inclusion of "style" also captures sections like "lifestyle." URLs and matching expressions were all processed as lower-case.

[6]While one might suggest that this classification captures "hard" and "soft" news, we do not want to suggest we are distinguishing between, say, news about a political scandal (soft) and coverage of a bill being introduced in the House of Representatives (hard).

[7]We remove non-English items and pre-process each summary by removing any HTML or English stop words like "a," "the," "and," and so on.

described above as $\mathbf{x}$ in our SVM, hence setting h to the identity function. By construction, SVMs impose a penalty on the coefficients, and here we use the ridge (L2) penalty. The ridge penalty is desirable under conditions of collinearity, because it shrinks highly predictive coefficients toward the same value (compared to the situation in which one coefficient dominates the rest in unpenalized models). This is especially appropriate to deal with the problem of polysemy in text classification, wherein multiple words mean the same thing.

This approach to generating labels has several advantages over relying on conventional human-coded labels. First, the large number of documents we model helps to overcome some of the sparsity in our features that would otherwise lead to highly variable predictions. This sparsity is due to the rich variety of words used in human language. It is complicated by the curse of dimensionality—as the number of features increases, the sampling density is diluted by $N^{\frac{1}{p}}$, meaning the variance of our predictions will increase with each new feature. This is particularly problematic when attempting to classify political content, in which the words used are even more highly variable and change based on current events.

Validation tests reveal that our classifier yields extremely accurate results. We assess the performance of our method using *cross-validation* (e.g., Hastie et al. 2009). Here we employ ten-fold cross-validation, which entails fitting a model using 9/10ths of the articles in our dataset, then assessing out-of-sample performance based on the remaining 1/10th. This is done ten times with each tenth of the articles, and the results are averaged. Based on ten-fold cross-validation, the classifier achieves accuracy of 0.946, precision of 0.933, and recall of 0.835.[8] This means that 95% of our out-of-sample classifications agreed with our labels; that given that a document is political, we identified it as such 84% of the time; and given that we made a prediction that a document was political, it was actually political 93% of the time. However, it is also desirable to examine randomly drawn examples of each document that our classifier predicts as political or not political from the full corpus of content (e.g., Grimmer and Stewart 2013). Table 1 below presents a sample of the text for both the political and non-political categories. As the classification metrics suggest, the classifier does extremely well, performing classifications as we would expect.[9]

[8] We achieved F1, the harmonic mean of precision and recall, of 0.881.

[9] It should be noted that this classifier is suitable for formal media prose and is inappropriate for the much harder problem of classifying informal discourse.

Table 1: Random Sample of 10 Articles per Class

Class Label	Sample of Text Summary
Political	In a rare glimpse into his tenure at Bain Capital, Mitt Romney writes about lessons learned and how they prepare him for the presidency...
Political	If you believe that the United States should legalize dogfighting because we allow humans to fight, fear not. You've got an ally in the...
Political	You don't need to be born in the U.S. to be president?
Political	The killings heighten worries about how the coalition troops can protect themselves while working at close quarters with their Afghan...
Political	Vice President Biden's handlers appear to be going to new lengths to keep their boss's antics and open to interpretation quips from ...
Political	WARSAW—A Mitt Romney aide told reporters to "shove it" Tuesday morning after the American press corps here shouted questions ...
Political	If Obama isn't...
Political	As reported Thursday by Katrina Trinko of National Review. If you are on the Obama fundraising e-mail list, you may have gotten a ...
Political	Over the past couple weeks some of Barack Obama's top fans have openly deserted him, expressing a commonly heard frustration...
Political	China steps up campaign against Ramadan fasting for Uighurs; experts fear backlash...
Non-Political	Chad Johnson and his wife Evelyn Lozada got into an altercation over the weekend, which ended wi...
Non-Political	Check my route, see more data and leave a comment by clicking the link above..
Non-Political	Puzzled Hearts Collection...
Non-Political	FOX Sports content...
Non-Political	Watch the best free live streaming tv and radio Fresh Radio 102.9 FM Windhoek 90.1 FM Oshakati ...
Non-Political	First Congregational United Church of Christ, 4600 Hamilton Blvd, welcomed a new pastor, the Re...
Non-Political	Bob Knight's analogy between handling stress and being raped, expressed during a documentary on ...
Non-Political	Do you have nightmares about going to school? Do you hate going to school because you know you ...
Non-Political	With fullback tight end David Johnson lost for the season due to a torn ACL, the Steelers have a...

Empirical Context

Before presenting our causal estimates, we note the striking relationship between exposure to political news on Facebook and our outcomes of interest, voter turnout and policy preferences. Figure 1 shows how individuals exposed to more socially shared political content are more likely to self-report that they had voted (exposure is overdispersed so we use a log scale to compress the measure). And Figure 2 shows that partisans exposed to more news are more likely to report more extreme policy preferences, while those who do not identify with a majority political party report more left-leaning preferences.

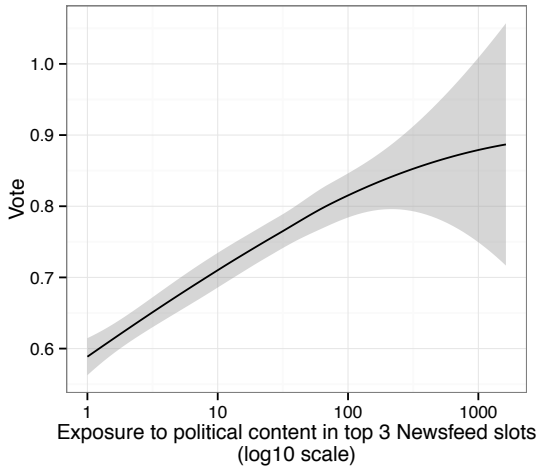

Figure 1: Individuals who tend to be exposed to more political news stories have greater self-reported levels of turnout. Line fit to a loess curve, band indicates standard errors.

Of course, an array of prior research shows that consumption of political media is associated with higher levels of political activism, voter turnout, and more extreme policy preferences. Yet it is difficult to determine whether media consumption itself can increase these outcomes, or whether this relationship arises due to the influence of some third factor such as general political interest and knowledge. In the present study, exposure is driven primarily by what an individual's friends share, not individual-level habits. Yet exposure is still highly correlated with turnout and policy preferences (Figures 1 and 2). This could be due to the fact that engagement with political content is correlated across friend networks (see Figure 3).

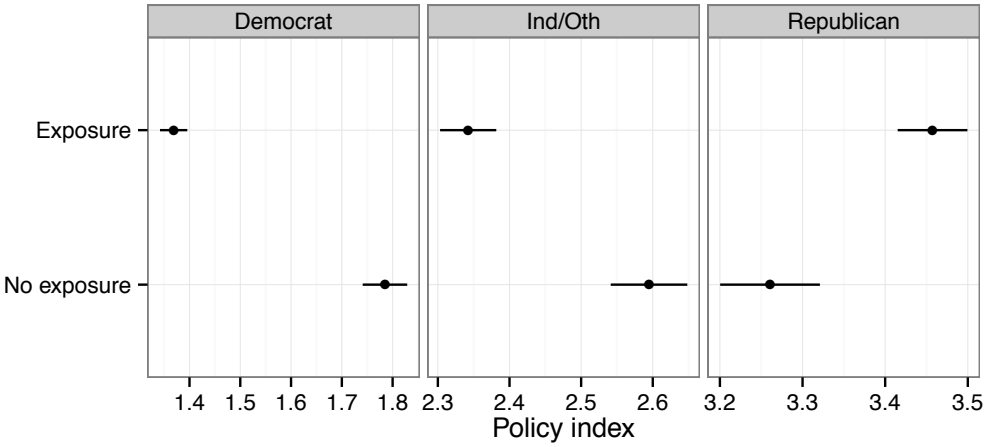

Figure 2: Observationally, individuals who get exposed to political news on Facebook hold more polarized policy preferences, while independents and non-identifiers hold slightly more left-leaning preferences (lower index values correspond to more left-leaning attitudes). Lines indicate standard errors.

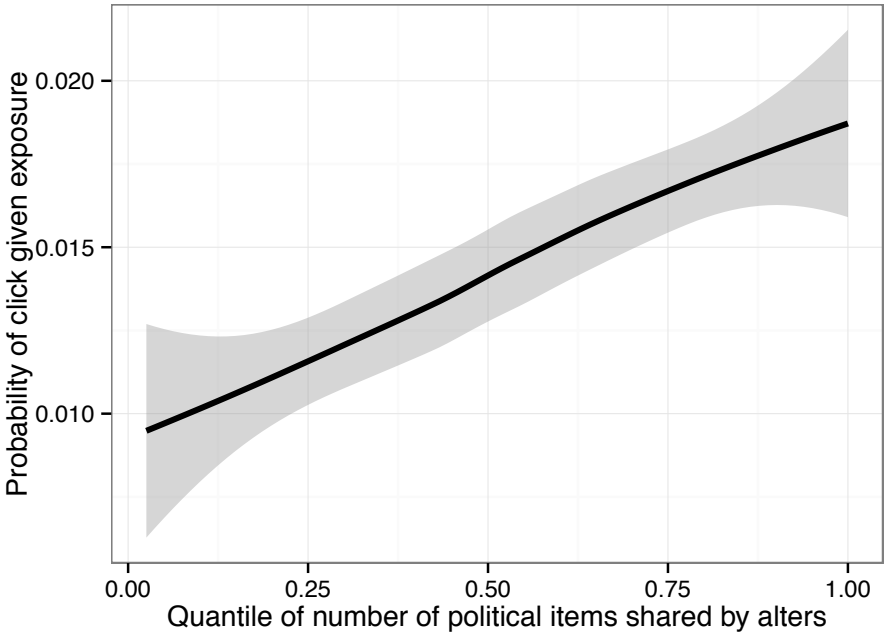

Figure 3: Individuals who have more friends sharing political content are more likely to click on any given political item appearing in their news feed. Plot shows click-through rate (CTR) as a function of the number of news items shared by friends. The horizontal axes is quantile-transformed to show how CTR varies over the entire population. Data is from August 1 - August 25, 2012 (before the test started). Band indicates standard error.

Finally, an individual's propensity to engage with political content on Facebook is related to the highest "slot" in which it appears in the news feed. Figure 4 shows that individuals click on political content that appears in the news feed in the first slot at nearly twice the rate as content that makes it only to the fourth slot. These order effects appear across a wide range of phenomena, from ballot order effects that can influence election outcomes (Koppell and Steen 2004) to lab-controlled news reading interfaces (Messing and Westwood 2012). Additionally, content is sorted based on a dynamic model that favors content with high interaction rates (among other factors), which may contribute to the observed relationship between click-through rate and position. Regardless, political content that appears in the top five slots comprises many of the shared links that individuals click to access the content at the originating website.

To examine the causal relationship between exposure to socially shared news on Facebook and these outcomes—turnout and policy preferences—we instrument the system as explained below.

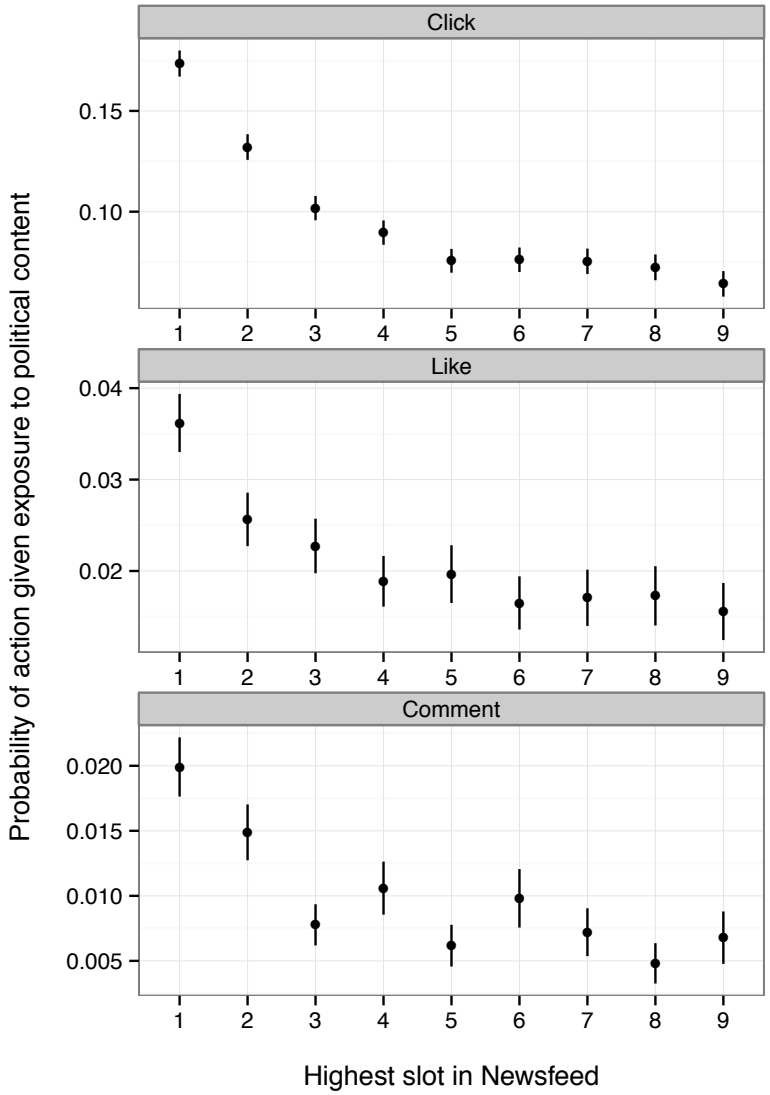

Figure 4: The likelihood of engaging with political content is highest for content that appears in the top slot in an individual's newsfeed, and decreases as the highest slot reached declines (control condition shown). Lines indicate standard errors.

Causal Analysis

Random Assignment

Beginning on August 26, 2012, 1.1 million U.S. Facebook users were randomly assigned to test a proposed change to Facebook's news feed that would emphasize socially shared news content on Facebook. The randomization scheme relied on simple random assignment and persisted across users for the entire time period in question. Randomization was orthogonal to key pre-treatment demographic and behavioral measures (full randomization check provided in the Appendix).

The Instrument

Users in this test group experienced a slightly modified version of the site that displayed links to political (and non-political) content from news media websites shared by friends in a more prominent position (Figure 5). This was achieved with a slight modification to the algorithm that ranks content for these users, such that items shared that contained links to news websites were more likely to appear in the news feed, and to appear toward the top. The difference for users in the test group is subtle, but occurred for a substantially longer time period than most previous media effects studies. We compare this group with 560 thousand Facebook users randomly allocated to a control group whose news feed would remain unmodified during the test period.

Outcome Measures

In the days following the 2012 election (November 7 to November 19), 75 percent of individuals over 18 in the test and control groups were asked to complete a survey that measured voter turnout, policy preferences, attitudes toward candidates and issues, political knowledge, and standard demographics. Individuals needed to log in to Facebook to see the invitation. The response rate (among all invitees, irrespective of whether they logged in) was 1 percent, yielding 8.3 thousand test group respondents and 4 thousand control respondents who completed at least partial responses.[10] The survey included standard questions on policy preferences, media consumption, reports of voting behavior, and feelings toward political parties and officials. We provide the full questionnaire and our analysis indicating a lack of differential attrition by group assignment due to survey non-response in the Appendix.

In our analysis we focus on reported turnout and policy preferences. We measure turnout

[10] Of these respondents, 48 percent answered at least 90 percent of the questions on the survey.

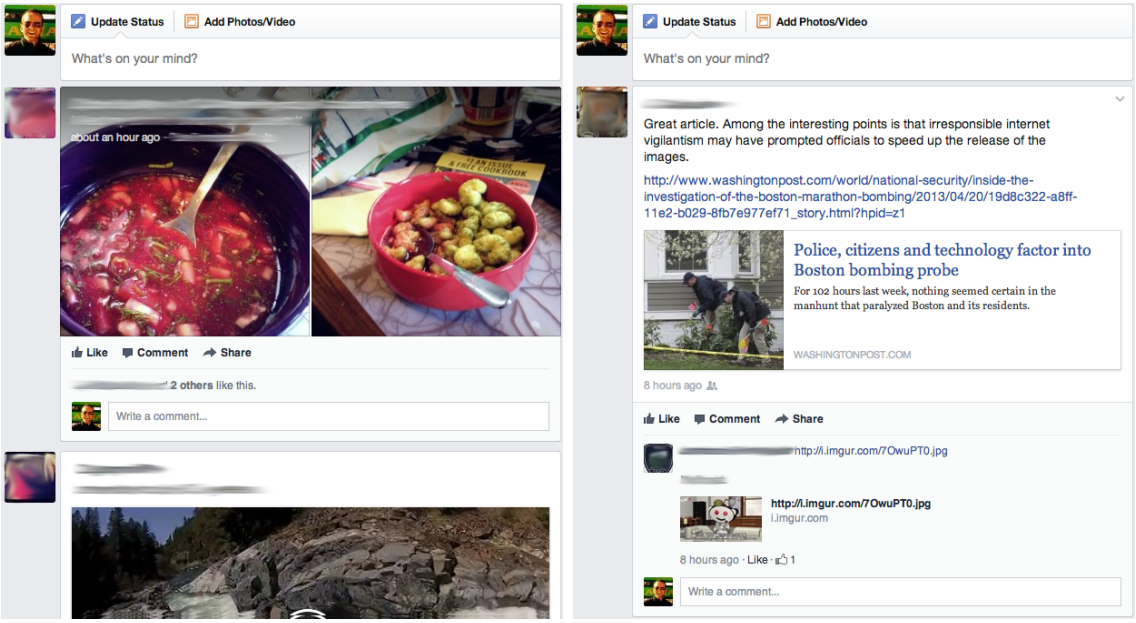

Figure 5: Illustrating the difference between test and control groups. The test group (right) had a higher chance of encountering news, civic affairs, and political content than the control group (left).

using the American National Election Study question wording. To capture policy preferences, we computed a six-point index of policy attitudes based on the extent to which respondents reported supporting the Patient Protection and Affordable Care Act ("Obamacare"), gun control, lower levels of federal spending, restricting access to abortion, and federal legalization of same sex marriage—respondents' answers to these questions were most strongly predictive of party affiliation (see Appendix for details).

Results

Instrumented Exposure to Political Content

Having used the method described above to classify content that appeared in the news feed as political during the 72-day test period, we examine the number of items classified as political content that made it to the top three slots in the news feed for respondents in the test and control groups. Figure 6 shows how much political content appeared in at least one of the top three positions in a respondent's news feed as a function of her friends' propensity to share political news content. (As we show above, the likelihood that a participant interacts with this content in some way is much greater when it is in one of these positions.) Figure 7 shows the number of political items that participants clicked on in the feed by test group.

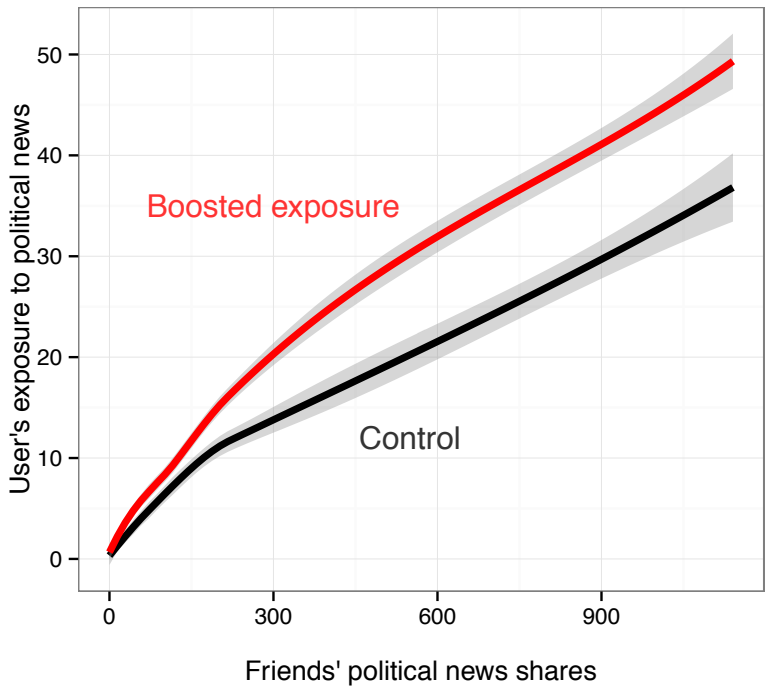

Figure 6: Ego's exposure to political content in the top slots in the news feed as a function of the aggregate count of all political content shared by ego's friends (alters), by test group. Those whose friends shared political content had a higher chance of encountering news, civic affairs, and political content than the control group. Lines are loess curves, bands are standard errors (data Winsorized at the 99th percentile).

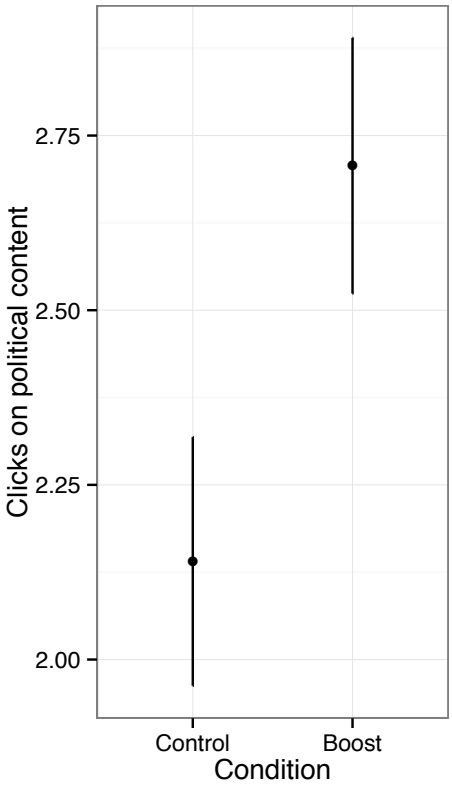

Figure 7: Those in the test group clicked through to more political stories during the study, indicating that additional political content was not merely ignored. Lines are standard errors.

Though we expect individuals to see more political news stories their friends share in the test group, the exact number participants saw was non-deterministic.[11] Hence, this setup naturally lends itself to an instrumental variables modeling strategy (Sovey and Green 2011). Our instrument—assignment to receiving the boosted levels of political content in news feed (Z)—satisfies the exclusion restriction, because assignment is random, and only affects outcomes through its effect on increased exposure to political news (D). And, as we will show below, it is correlated with the outcomes of interest, a relationship we refer to as the intent to treat (ITT) effect from here on.

We estimate these models using two stage least squares (2SLS), employing the standard correction in the second stage to calculate the covariance matrix correctly, as implemented in the R package AER.

Turnout and Political Salience

First, we compare self-reported turnout measure and our (post-treatment) political interest measure among those in test versus those in control to establish that our instrument Z affected our outcome Y (turnout). A simple comparison of the proportion of respondents who report voting reveals a significant intent to treat (ITT) effect on self-reported turnout—70.9 percent those in the boosting condition voted while 68.9 percent of those in the control condition report voting in the election (Wilcoxon rank sum test, $W = 10611494$, $p = 0.025$, two-sided). The test version of news feed significantly increased self-reported voter turnout.

[11] The exact number of news stories an individual was exposed to (D) may be correlated with their existing political attitudes and voting tendencies (as illustrated in previous sections).

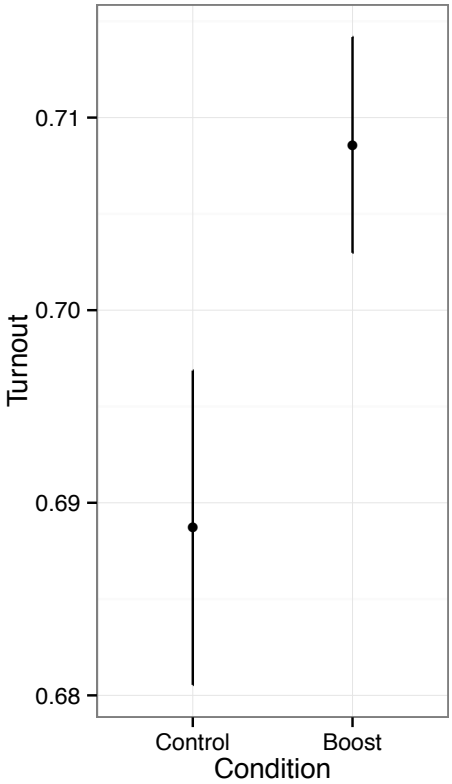

Figure 8: The ITT shows that the test version significantly increased self-reported voter turnout.

	Vote	Pol int
(Intercept)	0.605^{***}	2.509^{***}
	(0.047)	(0.109)
Exposure (log_2)	0.055^{*}	$0.101^{\dagger}$
	(0.027)	(0.061)
N	9807	8922

Table 2: Two-stage least squares estimates of the effect on self-reported turnout and political interest. The instrumental variable, Z, is random assignment to increased political content in news feed.

Our measure of political interest provides some evidence for why exogenous exposure to socially shared news increased turnout. If encountering political information shared by one's peers raises the social salience of politics, individuals should be expected to report higher levels of interest in politics. Indeed, political interest measured based on common ANES scales shows that individuals in the boosting condition ($M = 2.69$) were marginally more likely to report higher levels of political interest than those in control ($M = 2.66$, $W = 8565374$, $p = 0.097$).

We now present our instrumental variables regression of voting on exposure to political content. Because our exposure measure is heavily skewed, we transform it to a base 2 logarithmic scale. We also transform our political interest measure to a zero-one scale to make it easier to interpret. Our estimate of the effect for those who saw more political news in the test group, or the local average treatment effect (LATE, Table 2), shows that each two-fold increase in the number of political URLs to which users were exposed caused a 5.5 percentage point increase in the likelihood of voting. However, the LATE of exposure on political interest, while positive, is only marginally significant.

Policy preferences

In addition to reporting higher levels of political interest and turnout, non-partisans in the test group show stronger policy preferences (consistent with the literature on political knowledge and policy preferences, see e.g., Bartels 1996; Carpini and Keeter 1997). An ordinary least squares regression (containing indicators for Democrat and Republican political affiliation and everyday Facebook use) reveals an estimate of the ITT on the policy index equivalent to roughly one-sixth the effect of Democratic party affiliation, or one-thirteenth the effect of Republican party affiliation ($\beta = -0.11, SE = 0.04, T = -2.60, P = 0.009$). Whether this arose from argument frequency, argument quality, opinion activation, or some-

thing else remains unclear.[12] As Figure 9 shows, the effect is driven by non-partisans, who express slightly stronger support for liberal policies in the test group.[13]

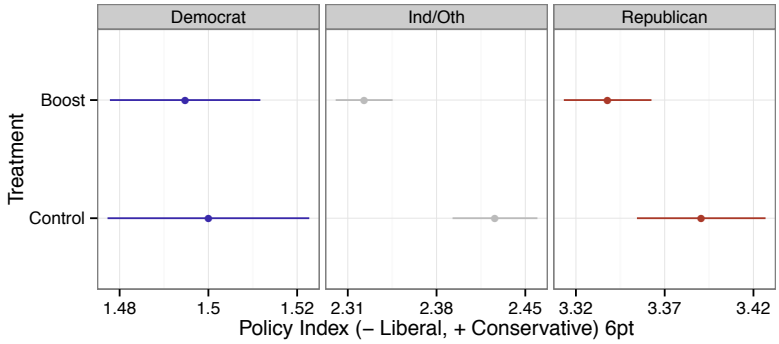

Figure 9: An examination of the ITT reveals stronger policy preferences among non-partisan identifiers in the test group. Whether this arose from increased knowledge, argument frequency, argument quality, opinion activation, or something else remains unclear.

[12]Because political party identification explains so much of the variation in policy preferences, it is unproductive to estimate this effect without the precision gained by conditioning on these variables.

[13]The effect of encountering additional socially shared news was strongest on social issues (abortion and gay marriage), but persisted when adding other policies to the index, including federal spending levels, health care, and gun control. We use the broader index in the above analysis.

	All	Independents	Democrats	Republicans
(Intercept)	2.535^{***}	2.773^{***}	1.519^{***}	3.632^{***}
	(0.101)	(0.200)	(0.138)	(0.259)
Exposure (log_2)	-0.128^{*}	-0.263^{*}	-0.013	-0.176
	(0.056)	(0.112)	(0.068)	(0.163)
Democrat	-0.783^{***}			
	(0.027)			
Republican	1.022^{***}			
	(0.029)			
N	6873	2194	2854	1825

Table 3: Two-stage least squares estimates of the effect on policy preferences. The policy index ranges from zero to five, where lower values indicate stronger support for liberal policies and higher values indicate stronger support for conservative policies. The instrumental variable, Z, is random assignment to the test group that saw additional political content.

Our instrumental variables analysis reveals the LATE in terms of exposure to political content. Again, our exposure measure is heavily skewed, so we transform it to a base two logarithmic scale. We also include indicators for whether the individual identifies as a Democrat or Republican as covariates in both stages of the model. As shown in Table 3, a two-fold increase in the number of political urls to which non-partisan users were exposed resulted in a -0.26 point shift on the index.

We cannot say whether these results mean that exposure to political information on Facebook reflect simple polarization in response to additional media exposure (e.g., Vallone et al. 1985, though we did not detect an effect for partisans), or whether respondents in the treatment condition provided responses that merely reflect knowing more about the issues, or perhaps something else. The latter interpretation is consistent with the literature on political information, which generally finds that informed individuals are more likely to hold stronger and more consistent policy preferences (Bartels 1996, 1988; Zaller 1992; Krosnick and Brannon 1993; Carpini and Keeter 1997) and is compatible with work showing that individuals who do not have attitudes on an issue often form them in the process of exposure to particular arguments and ways of framing different policies (see Druckman et al. 2012). This literature suggests that individuals in the treatment condition who did not identify as Democrats or Republicans could have formed more left-leaning policy preferences in the process of being exposed to left-leaning frames. Indeed, non-partisans were exposed to substantially more content that was shared by individuals identifying as liberal in their Facebook profiles than those identifying as conservative (see Appendix).

Discussion

In the context of a national election, additional exposure to socially shared political content increased self-reported turnout while generally strengthening policy preferences among nonpartisans. The encouragement design used here is especially appropriate to document the effects of real-world exposure. By subtly increasing exposure to socially shared media content for individuals randomly assigned to the test group, this design ensures internal validity while maintaining external and ecological validity. This is the first study of which we are aware to randomly assign individuals to encounter additional political content—socially shared or not—for such a long period of time, during an election season, and in which modern survey instruments are available to assess relevant outcomes. Yet, by obtaining precise measures of exposure to political media, we avoid the pitfalls of relying on self-reported measures of exposure to political information (thought to be plagued by issues related to faulty recall and social desirability bias, see Price and Zaller 1993; Prior 2009; Vavreck 2007).

This study is limited in that we can only study the effect of increasing exposure to news on Facebook shared by peers for those who could be exposed to more news. Some individual's friends simply do not share a sufficient amount of political content such that additional news could be shown, and, as we find in the empirical context section above, this group has a lower propensity to vote and holds less extreme policy preferences.[14] This is a natural limitation of the local average treatment effect identified by our instrumental variables analysis (Sovey and Green 2011).

Nevertheless, this work helps address long-standing questions about how increases in civic engagement among individuals' peer groups can influence their opinions and behavior (Berelson et al. 1954; Katz and Lazarsfeld 1955; Delli Carpini 2000; Shah et al. 2001; Bennet 2008; Gerber et al. 2008; Pasek et al. 2009; Gil de Zúñiga et al. 2012), and about how social factors and algorithmic aspects of media distribution shape our information environment (Lewin 1947; Lazarsfeld et al. 1948; Schramm 1949; Freedman and Sears 1965; Pariser 2011; Bakshy et al. 2012; Muchnik et al. 2013).

These findings underscore the wide variance in exposure to political content in social media, which depend on the behavior of one's social contacts. Of course, this phenomenon is not new (having been discussed at length in, for example, Sears and Freedman 1967). Yet, the massive increase in horizontal channels of distribution (i.e., social media and email) in recent years suggests this dimension of information distribution is more important than in previous times. Our work shows some of the consequences of this kind of exposure to

[14] Of course, the treatment may have caused friends of participants in the control condition to see additional news via re-sharing, though the low rate of re-sharing mitigates this concern, and if anything this would serve to make our estimates of the effect size more, not less, conservative.

political content for political preferences and behavior. The shift in media distribution and consumption in favor of social channels means that this study constitutes an important step in understanding the implications for political preferences and behavior. This shift also means that this study should, if anything, grow in terms of relevance and external and ecological validity as the prominence of social media for these purposes continues to rise.

The fact that these effects depend so heavily on exposure to what our friends share means that factors related to social influence—tie strength, status, and perceived expertise—may interact with content to drive attention and ultimately influence outcomes relevant to politics, hypotheses suggested by our exploratory analyses of heterogeneous treatment effects in this data, see Appendix IV. Indeed, this work supports the hypothesis that exposure to elite discourse is increasingly dependent on endogenous social factors (Bennett and Iyengar 2008), and suggests that as the prevalence of socially shared media increases, agendas should be expected to grow increasingly varied and individualized. Future research should explore how these social factors impact exposure and interact with content to affect political preferences and behavior.

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Appendix

Randomization check

Randomization was orthogonal to friends' propensity to share political content, as well as political affiliation and other relevant demographics. We take self-reported political affiliation and scale it from -1 (Liberal/Democrat) to 1 (Conservative/Republican, full details available in the Supplementary Information), then test for differences between the boosting treatment and the control using the Wilcoxon Rank Sum test. Differences between the boosting group ($\mu = .017$) and control ($\mu = .016$) were not significant ($W = 891690336$, $p = .47$, two-sided). Likewise, the proportion of males to females did not differ significantly in the boosting treatment ($\mu = .427$) versus control ($\mu = .427$), ($W = 63876961992$, $p = .58$, two-sided). In addition, any difference in age between the boosting group ($\mu = 34.9$) and control ($\mu = 34.9$) was not significant ($T(498641.9) = 0.46, p = .64$, two-sided).

Survey Question Wording

How would you describe your political affiliation?
-Democrat
-Republican
-Independent

In any election, some people are not able to vote because they are sick or busy or have some other reason, and others do not want to vote. Did you vote in the election held on Tuesday, November 6, 2012?
-Yes, I voted
-No, I did not vote
-I usually vote but didn't this time
-I thought about voting this time but didn't
-I cannot remember

Which of the following best describes how often you follow what's going on in government?
-Hardly at all
-Sometimes
-Often

-Most of the time

On a scale from 0 (coldest) to 100 (warmest) how do you feel about the following people?
-Barack Obama
-Mitt Romney
-Democrats
-Republicans

Survey Question Wording

Do you support or oppose the following policies (6pt, strongly oppose - Strongly support)
-The Affordable Care Act ("Obamacare") -Gun control
-Lower taxes
-Lower levels of government spending
-"Cap and trade" (bill to limit carbon emissions)
-Stronger controls on campaign spending
-DREAM Act (path to citizenship for young undocumented immigrants)
-Mandating insurance companies to provide contraception
-Tighter legal restrictions on abortion
-Federal legalization of same-sex marriage

Policy Index

Table 4 shows the regression of partisan affiliation (measured as a 3 point scale) on each question response (6 point, strongly oppose - strongly support). We include predictors that are significant at $P < 0.01$ in our index of partisan policy preferences.

Table 4: OLS model predicting partisan affiliation (measured on a 3 point scale) based on responses to questions about the level of support (6pt, strongly oppose - strongly support) for each policy.

(Intercept)	0.810***
	(0.050)
Obama Care	−0.219***
	(0.005)
Gun Control	−0.030***
	(0.005)
Lower Taxes	−0.012*
	(0.006)
Lower Government Spending	0.043***
	(0.007)
Cap & Trade	−0.020**
	(0.006)
Campaign Spending Regulation	0.010
	(0.006)
DREAM Act	−0.021***
	(0.005)
Mandatory Contraception Provision for Ins Co.	−0.021***
	(0.005)
Tighter Abortion Restrictions	0.045***
	(0.005)
Same Sex Marriage	−0.029***
	(0.005)
R-squared	0.522
adj. R-squared	0.521
N	6283

Attrition

The population that we study are the set of individuals who are assigned to the test or control group, log in to Facebook, and take the survey. Because we only observe outcomes (policy preferences and voter turnout) for individuals who responded to the survey, one threat to validity is differential attrition: that individuals who we observe at the end of the experiment (those that fill out the survey) differ between conditions in important ways.

We designed our survey recruitment and question wording to minimize the salience of politics at the time of deciding to respond. Respondents were initially recruited into the survey with the prompt "We'd like to hear from you. Please take 2-3 minutes to share your thoughts and opinions." Furthermore, our measures of turnout and political preferences using ANES wording were issued after a standard battery of demographic questions.

Because so many Facebook users self-report their political affiliation and demographics, our study allows for an extensive assessment of differential attrition due to survey non-response by subgroup. We model response rates as a binary outcome and condition on various demographic to examine the predictors of survey response.

Table 5 shows that those in the test condition are not significantly more likely to respond to the survey. Those who respond to the survey are substantially more likely to use Facebook, to report their political affiliation, to be embedded in networks that report political affiliation, and skew slightly left. However, we do not observe differential attrition in any of those subgroups. A likelihood ratio test between model specifications containing these subgroups (2) and including the treatment indicator and all treatment interactions (3), suggests that (3) does not yield significantly more explanatory power ($\chi^2(11) = 8.74$, P = 0.35).[15]

[15]Treating age as an integer variable does not change these results.

Table 5: Predictors of survey response, logistic regression coefficients. Individuals who are more active on Facebook and individuals who are more involved in politics were more likely to respond to the survey. Results do not indicate differential attrition by condition.

	(1)	(2)	(3)
(Intercept)	-5.006^{***}	-5.489^{***}	-5.505^{***}
	(0.016)	(0.022)	(0.039)
T	0.033		0.024
	(0.019)		(0.047)
Age 25-34		-0.247^{***}	-0.258^{***}
		(0.024)	(0.041)
Age 35-44		-0.211^{***}	-0.248^{***}
		(0.027)	(0.047)
Age 45-54		-0.104^{***}	-0.141^{**}
		(0.030)	(0.053)
Age 55-64		-0.234^{***}	-0.250^{***}
		(0.039)	(0.068)
Age 65 +		-0.331^{***}	-0.450^{***}
		(0.050)	(0.091)
Male		-0.188^{***}	-0.191^{***}
		(0.019)	(0.033)
Everyday FB user		1.640^{***}	1.660^{***}
		(0.019)	(0.034)
Conservative Affil.		0.632^{***}	0.587^{***}
		(0.036)	(0.064)
Liberal Affil.		0.776^{***}	0.858^{***}
		(0.033)	(0.056)
T × Age 25-34			0.016
			(0.050)
T × Age 35-44			0.055
			(0.057)
T × Age 45-54			0.055

			(0.064)
T × Age 55-64			0.025
			(0.083)
T × Age 65 +			0.173
			(0.109)
T × Male			0.006
			(0.040)
T × Everyday FB user			−0.029
			(0.041)
T × Conservative Affil.			0.066
			(0.077)
T × Liberal Affil.			−0.123
			(0.070)
Log-likelihood	−74540.236	−68871.163	−68865.607
Deviance	149080.472	137742.325	137731.215
AIC	149084.472	137762.325	137771.215
BIC	149109.311	137886.163	138018.889
N	1829907	1765128	1765128

Exposure to Content by Sharer's Partisanship

Above, we show that individuals in our test group who do not affiliate with one group of partisans reported stronger and more consistent support for left-leaning policies than the non-partisans in the control group, and we speculated in the discussion that this could be due to exposure to content that was framed in a particular way as they formed opinions on issues. One indication of partisan framing is whether an individual who shares an article identifies as a partisan herself. After all, sharing constitute an endorsement (of the importance of the piece if not the way facts and opinions are presented therein). In Figure 10, we present the average proportion of political items appearing in the top 3 slots in participants' newsfeeds by the ideology of the sharing friends. Non-partisans tended to be exposed to a higher proportion of content shared individuals with a left-leaning ideology than a right-leaning ideology, consistent with the framing-opinion formation explanation.

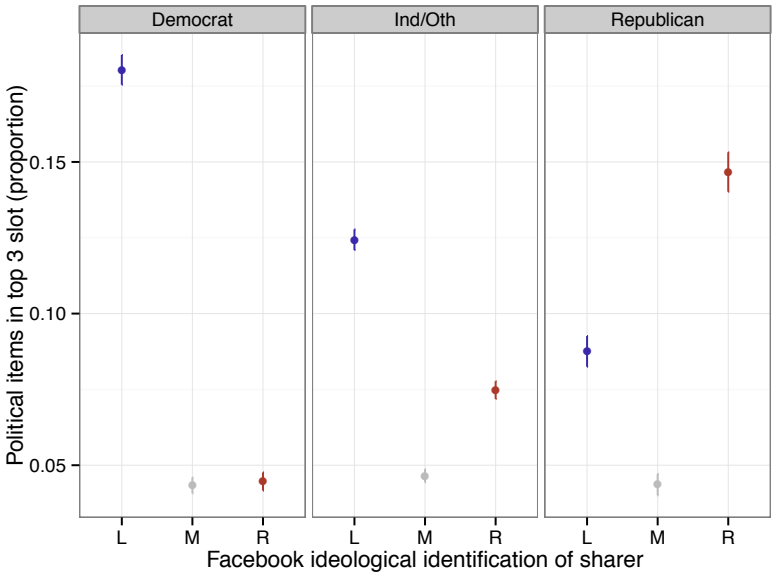

Figure 10: Non-partisans were exposed to a higher proportion of content shared individuals with a left-leaning reported ideology than a right-leaning reported ideology. Content appearing in participants' newsfeeds by ideology of friends sharing. 3-point left, middle, right ideological categories based on a mapping from the top 500 self-reported "politics" designations from users profiles.